COMPUTER SYSTEMS AND PLATFORM TECHNOLOGIES

IT0047

FEU ALABANG FEU DILIMAN FEU TECH

Technology Driven by Innovation

MODULE 2.1

Review on Computer Concepts: How Computers Work

At the end of the lesson, the student should be able to :

- **Define Computer**
- **Identify elements of a computer system**
- **Identify the input, output and storage devices**

Computer

A ***Computer*** is an electronic device that is capable of accepting data/information, performs arithmetic and logical operation and supplies the result in an acceptable form.

Hardware

Tangible parts of a computer system, includes:

Memory
CPU
Input devices
Output devices
Buses

Input Devices

Used to input instructions into the computer system.

Keyboard
Mouse
Joystick
Scanner
Touch Panel
Microphone

Output Devices

Used to give an output of what the computer executes.

Monitor
Printer
Projectors
Speakers

Software

These are programs that run the computer, designed to perform a particular type of task.

How Computers Work

Operating System

It is the software that makes the computer work.

Peopleware

Peopleware refers to anything that has to do with the role of people in the development of computer software and hardware systems.

Computer Memory

It stores data and program instructions.

RAM – Random Access Memory
which can be read or changed by the user or computer.

ROM – Read Only Memory
which can be read by the computer but not altered in any way.

ROM Memory

PROM - Programmable Read Only Memory

Information is written to a PROM chip after it is manufactured. A PROM chip cannot be erased or re-written.

ROM Memory

EPROM - Erasable Programmable Read Only Memory

Information is written to an EPROM chip after it is manufactured. An EPROM chip can be erased with exposure to UV light.

ROM Memory

EEPROM - Electrically Erasable Programmable Read Only Memory

Information is written to an EEPROM chip after it is manufactured. EEPROM chips are also called Flash ROMs. An EEPROM chip can be erased and re-written without having to remove the chip from the computer.

Central Processing Unit (CPU)

Information from an input device or from the computer's memory is communicated via the bus to the ***central processing unit (CPU), which is the part of the computer that translates commands and runs programs.***

Functional Unit of CPU

Arithmetic and Logic Unit (ALU)

is where the execution of most operations takes place. It contains a number of high-speed storage elements called ***registers.***

Control Unit (CU)

is the nerve center of a computer. It sends control signals to other units and senses their state.

MODULE 2.2

Review on Computer Concepts: Programming Language

At the end of the lesson, the student should be able to :

- **Define Programming Language**
- **Identify the types of programming language**
- **Identify the advantages of using High Level and Low Level Languages**

Programming Languages

- contain the series of commands that create software, all other programming languages must be converted to machine code for them to be understood because a CPU has a limited set of instructions known as machine code that it is capable of understanding.
- ***Machine Language***
  - A program is a sequence of extremely simple instructions known as ***machine code ,*** Machine code instructions are binary

    which are not easy to understand

High-Level Language

- This programming languages were developed because of the difficulty of programming using assembly languages
- High-level languages are easier to use than machine and assembly languages because their commands are closer to natural human language
- A compiler turns a high-level program into a CPU-specific machine language.

Assembly Language

- A programming language that uses easy-to-remember commands or ***mnemonics in place of binary patterns in order to make the task of programming easier***
- Once an assembly-language program is written, it is converted to a machine-language program by another program called an ***assembler***
- Both machine and assembly languages are ***low-level programming languages***

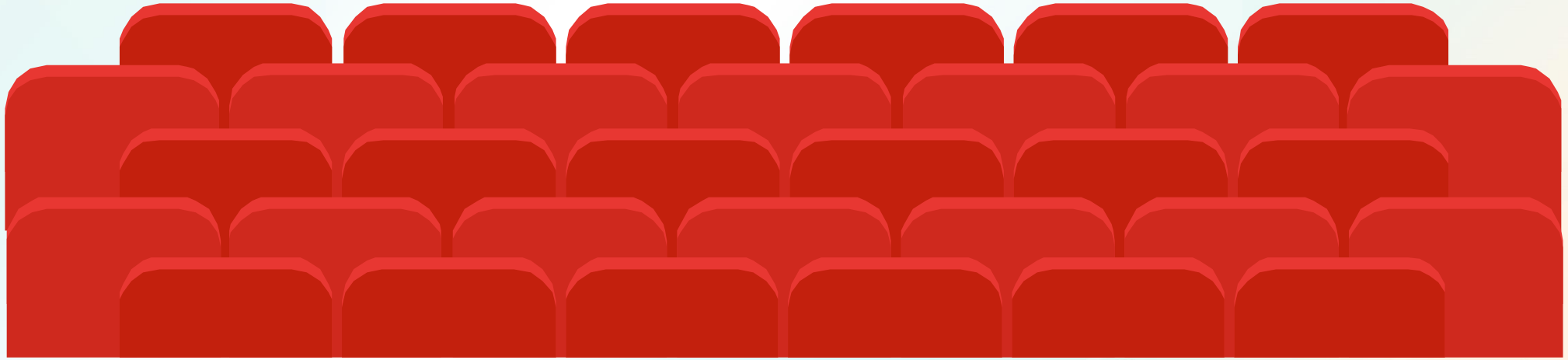
Advantages of using high-level languages over low-level languages:

- *Easy to Learn*
- *Predefined Functions*
- *Portability*

Advantages of using low-level languages over high-level languages:

- *Compact Code*
- *Speed*
- *Flexibility*

Q&A SESSION

ASK ANY QUESTION RELATED TO OUR TOPIC FOR TODAY.

THANK YOU